

CITS3002: Lab2-Solution

Exercise 1: Considering **1001** is used as the divisor in the CRC error detection:

a) Find data delivered to the receiver when the sender transmits **1011001**, assuming that no error occurs.

Answer:

Since the divisor has 4 bits. Hence, $4-1=3$ bits must be appended to the data bits before performing the CRC calculation.

$$\begin{array}{r} 1001 \overline{) 1011001000} \\ \underline{1001} \\ 001000 \\ \underline{1001} \\ 0001100 \\ \underline{1001} \\ 01010 \\ \underline{1001} \\ 0011 \end{array}$$

The CRC remainder is 011, so it is appended to the original data bits. Therefore, the transmitted bit pattern is $1011001000 + 011 = \mathbf{1011001011}$

b) If the receiver obtains 1100100101, does it detect any error? Explain why.

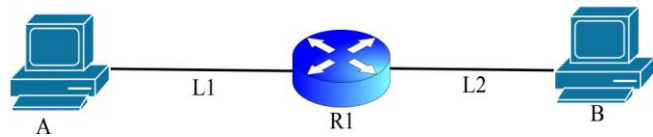
Answer:

$$\begin{array}{r} 1001 \overline{) 1100100101} \\ \underline{1001} \\ 01011 \\ \underline{1001} \\ 001000 \\ \underline{1001} \\ 0001101 \\ \underline{1001} \\ 0100 \end{array}$$

Remainder is not zero, so the received data is incorrect.

Exercise 2: Consider two hosts, A and B, are connected by two links (L1, L2). Suppose node A sends a packet with the size of 32 bytes to node B. Answer the following questions (queueing delays can be ignored):

Bandwidth of L1= 1000 bps
 Bandwidth of L2= 2000 bps
 Length of L1= 10 m
 Length of L2= 20 m
 Propagation speed of links= 2×10^8 (m/s)



a) What is the end-to-end delay?

Answer:

Transmission Delay of L1 = $(32 \times 8) / 1000 = 0.256$ seconds = 256 ms

Transmission Delay of L2 = $(32 \times 8) / 2000 = 0.128$ seconds = 128 ms

Propagation Delay of L1 = $10 / (2 \times 10^8) = 5 \times 10^{-8} \text{ s} = 0.00005 \times 10^{-3} \text{ s} = 0.00005 \text{ ms}$

Propagation Delay of L2 = $20 / (2 \times 10^8) = 1 \times 10^{-7} \text{ s} = 0.0001 \times 10^{-3} \text{ s} = 0.0001 \text{ ms}$

Delay of Link1 = Transmission Delay of L1 + Propagation Delay of L1

Delay of Link2 = Transmission Delay of L2 + Propagation Delay of L2

Total Delay = Delay of Link1 + Delay of Link2

256 ms + 0.00005 ms + 128 ms + 0.0001 ms = 384.00015 ms

b) From the moment Router R1 receives the packet, how long does it take for the first bit to reach Host B?

Answer:

When router R1 receives the packet, it must spend transmission time for one bit to push the first bit onto link L2. After the first bit is placed on the link, it requires the propagation delay of L2 to travel across the link and reach host B.

Transmission delay for one bit (from R to B) = $1 / 2000 = 0.5 \text{ ms}$

Propagation Delay of L2 = $20 / (2 \times 10^8) = 0.0001 \text{ ms}$

Total delay = 0.5 ms + 0.0001 ms = 0.5001 ms

Exercise 3: Hamming code is used for error detection and correction. The received bit pattern is 110110001110. Assume the original data length is 8 bits.

a) Determine whether a single-bit error has occurred.

Answer:

1	1	0	1	1	0	0	0	1	1	1	0
B12	B11	B10	B9	P8	B7	B6	B5	P4	B3	P2	P1

$P1 = B3 \text{ XOR } B5 \text{ XOR } B7 \text{ XOR } B9 \text{ XOR } B11 = 1$

$P2 = B3 \text{ XOR } B6 \text{ XOR } B7 \text{ XOR } B10 \text{ XOR } B11 = 0$

$P4 = B5 \text{ XOR } B6 \text{ XOR } B7 \text{ XOR } B12 = 1$

$P8 = B9 \text{ XOR } B10 \text{ XOR } B11 \text{ XOR } B12 = 1$

The computed values for P1 and P2 are different with received values. Hence, a single-bit error occurred.

b) If an error exists, identify the position of the erroneous bit.

$1 + 2 = 3$, so the error is at position 3 (bit B3).

c) Provide the corrected bit pattern.

B3 must be changed to zero, as a result we need to compute the parity again (by considering zero value for B3):

$$P1 = B3 \text{ XOR } B5 \text{ XOR } B7 \text{ XOR } B9 \text{ XOR } B11 = 0$$

$$P2 = B3 \text{ XOR } B6 \text{ XOR } B7 \text{ XOR } B10 \text{ XOR } B11 = 1$$

1	1	0	1	1	0	0	0	1	0	1	0
B12	B11	B10	B9	P8	B7	B6	B5	P4	B3	P2	P1